Steady State Cost in Competing Aggregation-Fragmentation Dynamics

Fitzwilliam James Keenan-Koch[1,*]

[1]Vermont Complex Systems Institute, University of Vermont, Burlington, VT, USA
[*]fkeenank@uvm.edu

ABSTRACT

Aggregation and fragmentation among polymers has long been an area of active research and utility in physical chemistry. Drawing inspiration from the dynamics of social formulation, we make a preliminary effort at optimization among various potential fragmentation kernels to regulate the probability of gelation. This work has relevance for any work formalizing the formulation, dissolving, and splitting of groups of abstract individuals. Social groups are of particular interest, from musical subcultures to the regulation of corporate consolidation.

1 Introduction

Things fall apart, Keats wrote. Of course, they also come together.

Whether *homo economicus* or *homo sociologicus*, the coordinating and organizational capabilities of the human race are broadly agreed upon as one of our key defining features. W

We attempt

The Introduction section can use referenced texts, for example[1], which calls references listed as bibitems in the `sample.bib` bibliography file. The introduction should expand on the background of the work (some overlap with the Abstract is acceptable) and explain your interests in the problem. The introduction should not include subheadings.

Aggregation is generally the transition

$$A_i + A_j \xrightarrow{K_{ij}} A_{i+j}$$

where: - A is a cluster - i and j are masses - K_{ij} is the rate of aggregation between clusters of mass i and mass j into a clusters of mass $i+j$ - K, then, is an infinite symmetric matrix $K_{i,j} = K_{j,i}$

The master equations for which are:

$$\frac{\mathrm{d}c_k}{\mathrm{d}t} = \frac{1}{2}\sum_{i+j=k} K_{ij}c_ic_j - c_k \sum_{i\geq 1} K_{ik}c_i \tag{1}$$

$$= \frac{1}{2}\sum_{i,j\leq 1} K_{ij}c_ic_j \left[\delta_{i+j,k} - \delta_{i,k} - \delta_{j,k}\right] \tag{2}$$

where δ is the Kronecker delta.

We propose additional **disaggregation** based on the general idea

$$A_k \xrightarrow{L_{ki}} A_i + A_{k-i}$$

where L_{ki} is the rate of disaggregation from clusters of mass k to clusters of mass $i, k-i$ such that L is **not** a symmetric matrix, as L_{31} is potentially valued and L_{13} is certainly 0. And with master equations:

$$\frac{\mathrm{d}c_k}{\mathrm{d}t} = \sum_{i\geq 1} [L_{ik}c_i(1+\delta_{i,2k}) - L_{ki}c_i]$$

where the Kronecker delta allows us to double count when the cluster splits into two equal clusters of size k.

2 Methods: A model of something

Up to three levels of **subheadings** are permitted. Subheadings should not be numbered (this is done by using an asterisk after any `subsection` or `subsubsection` command.

The problem

The first section should be a clear definition of the problem statement. That is, what is the system to be modelled and (if applicable) what specific research questions are you after.

This is example of more text under a subsection. Bulleted lists may be used where appropriate, e.g.

- First item
- Second item

The model

A clear model, specifying the modeling approaches, assumptions, parameters, objectives (what it should be used for or not). Consider using a model card once we revisit the concept in class.

Third-level section

Topical subheadings are allowed so feel free to change the `subsection` or `subsubsection` titles. This is an example of a `subsubsection`.

3 Results

The results can contain subheadings as desired and can discuss the results directly.

Figure 1. Legend (350 words max). Example legend text for an example figure. Figures should be in pdf format unless it is a photo or archival image not in vector graphics, as is the case here (in which case jpg and png are acceptable).

4 Discussion

The Discussion should be succinct. It typically involve any big picture discussions not covered in Sec. 3, i.e., the Results section. This was an example of a section reference using the `label` and `ref` commands. Figures and tables can also be referenced in LaTeX using the `label` and `ref` commands, e.g. Fig. 1 and Tab. 2.

This discussion can therefore mention scientific consequences of the results, ideas for future work, stress approximations or simplifications that limit the applicability of the model, or discuss things you tried that did not work. The Discussion section must not contain subheadings.

References

1. Figueredo, A. J. & Wolf, P. S. A. Assortative pairing and life history strategy – a cross-cultural study. *Hum. Nat.* **20**, 317–330, DOI: https://doi.org/10.1007/s12110-009-9068-2 (2009).

Acknowledgements

Acknowledgements should be brief, and should not include any effusive comments. Colleagues from the class (or beyond) who helped should be acknowledged here. Funding information, if any, may also be acknowledged.

Additional information

Data availability statement: if applicable.
Code availability statement: mandatory.
Competing interests: if any.

Table 1. Model Card

1. Model Details	
The problem	
Modeling approach	
Degrees of freedom	
Number of parameters	
Implementation language	
Relevant publications	
License	
Contact information	
2. Intended Use	
Primary intended uses	
If forecasting, validation metrics	
If applicable, data incorporated	
Primary users	
Out-of-scope uses	
3. Key assumptions	
Population size	
Contact structure	
Time structure and memory effects	
Other	
4. Metrics	
Computational complexity	
Variation approaches	
8. Ethical Considerations	
Potential risks and harms	
Bias and fairness concerns	
Mitigation strategies	
Impact on stakeholders	
9. Suggested next steps	
Issues with implementations	
Recommendations for use	
Future work	

Condition	n	p
A	5	0.1
B	10	0.01

Table 2. Legend (350 words max). Example legend text for an example table.